testbook PASS PRO MAX

General Knowledge

Indian Geography

STUDY NOTES

Drainage System: Himalayan River System

Drainage System: Himalayan River System

The **Himalayan Drainage System** is one of the most **extensive** and **crucial drainage networks** in the world. This complex system of **rivers**, originating from the **glaciers** and **snowfields** of the **Himalayas**, has been instrumental in shaping the **geography**, **ecology**, and **socio-economic fabric** of the **Indian subcontinent**.

About the Himalayan Drainage System

- The **Himalayan River System** primarily comprises the **Indus**, **Ganga**, and **Brahmaputra** river basins. These rivers are **perennial**, receiving water from both **snowmelt** and **precipitation**.
- The rivers flow through **deep gorges** formed by **erosional processes** alongside the **uplift of the Himalayas**.
- In the **mountainous course**, they create **U-shaped valleys**, **V-shaped valleys**, **rapids**, and **waterfalls**.
- As they reach the **plains**, these rivers form various **depositional features**, including **flat valleys**, **meanders**, **ox-bow lakes**, **flood plains**, **braided channels**, and **deltas** at their mouths.
- In the **Himalayan region**, the rivers follow a **tortuous course**, but in the **plains**, they show a tendency to **meander** and often **shift their courses**.
- The **Kosi River**, known as the **'Sorrow of Bihar'**, is notorious for its **unstable course** and susceptibility to significant **erosion**, which increases its **sediment load**, causing it to **change its course frequently**.

Evolution of the Himalayan Drainage System

- Geologists have differing views on the **evolution** of the **Himalayan rivers**. It is believed that a major river, called **Shiwalik** or **Indo-Brahma**, once traversed the entire length of the **Himalayas**, from **Assam** to **Punjab**, discharging into the **Gulf of Sindh** during the **Miocene period** (approximately 5-24 million years ago).
- Evidence of this ancient river includes **lacustrine origins** and **alluvial deposits**.
- Over time, this **Indo-Brahma River** divided into three major **drainage systems**:
  - **Indus** and its tributaries in the **Western part**.
  - **Ganga** and its **Himalayan tributaries** in the **Central part**.
  - **Brahmaputra** in **Assam** and its tributaries in the **Eastern part**.
- This division is thought to have occurred due to the **Pleistocene upheaval**, including the **uplift of the Potwar Plateau** (**Delhi Ridge**) in the west, which created a **water divide** between the **Indus** and **Ganga systems**. Similarly, the **Malda gap** between the **Rajmahal hills** and the **Meghalaya plateau** redirected the **Ganga** and **Brahmaputra systems** toward the **Bay of Bengal**.

Features of the Himalayan River System

- **Young Drainage Pattern**: The rivers are **young** and exhibit a **dendritic** or **trellis drainage pattern** with numerous **tributaries** joining the main river.
- **Seasonal Flow**: The rivers experience significant variations in **water flow** throughout the year, with **peak flows** during the **summer** due to **snowmelt** and **monsoon rains**.
- **Steep Gradients and V-Shaped Valleys**: The rivers have **steep gradients**, originating in **high-altitude regions**, resulting in **V-shaped valleys** and **rapid flow rates**.
- **Meandering Courses and Oxbow Lakes**: In the **plains**, the rivers exhibit **meandering courses**, forming **oxbow lakes**, **floodplains**, and **braided channels**.
- **Sediment-laden Rivers**: Due to their **steep gradients** and **erosive power**, **Himalayan rivers** carry large amounts of **sediment**, which they deposit in the plains, creating **fertile alluvial soils** in areas like the **Indo-Gangetic Plain**.

Indus River System

The **Indus River System**, a vital part of the **Himalayan drainage system**, plays an essential role in sustaining **ecosystems** and **human settlements** in the **Indian subcontinent**. This system is crucial for the **cultural**, **agricultural**, and **economic landscapes** of the region.

About the Indus River System

- The **Indus River System** is one of the three major **river basins** within the **Himalayan drainage system**.
- It flows through the **western part** of the **Indian subcontinent**, carving **deep gorges** and supporting a variety of **ecosystems** along its path.
- The **Indus River**, at over **3,000 kilometers**, is the **longest river in Pakistan** and one of the **longest in Asia**.

Origin of the Indus River

- The **Indus River** originates from a **glacier** near **Bokhar Chu** in the **Tibetan region**, in the **Kailash Mountain range** near **Mansarovar Lake**.
- It flows **northwest**, entering the **Ladakh region** of **India** at **Demchok**.
- The river runs between the **Karakoram** and **Ladakh ranges**.
- In **Tibet**, it is known as **'Singi Khamban'**, or the **Lion's Mouth**.

Course of the Indus River

- The **Indus River** is joined by the **Zaskar River** at **Leh** and later by the **Shyok River**.
- Near **Mithankot**, it receives water from the five eastern tributaries: **Jhelum**, **Chenab**, **Ravi**, **Beas**, and **Satluj**, which converge to form the **Panchnad**.
- The river continues through **Sindh Province**, gathering **sediment** to form the **Indus River delta** before draining into the **Arabian Sea** near **Karachi**.
- **Note**:
  - The **blind Indus River Dolphin**, a sub-species of dolphin, is found exclusively in the **Indus River**.
  - The **Indus** flows through **India** only in the **Leh district** of **Ladakh**.

Tributaries of the Indus River

The **Indus River** is fed by several **left-bank** and **right-bank tributaries**.

Left-bank Tributaries

The **left-bank tributaries** of the **Indus River** include:

- **Zaskar River**
- **Suru River**
- **Soan River**
- **Jhelum River**
- **Chenab River**
- **Ravi River**
- **Beas River**
- **Satluj River**
- **Panjnad River**

Important Left-bank Tributaries:

- **Zanskar River**: An important tributary with sparse **human settlements** along its course.
- **Chenab River**:
  - Originates from **Bara Lacha Pass** in the **Lahul-Spiti region**.
  - Formed by the confluence of the **Chandra** and **Bhaga rivers**.
  - Known as **Chandrabhaga** in its upper reaches.

- Flows through the **Jammu region** and into the **plains of Punjab**.
- **Jhelum River**:
  - Originates from a **spring** at **Verinag** at the foot of the **Pir Panjal** in the **Kashmir Valley**.
  - Its largest tributary, the **Kishanganga (Neelum) River**, joins it before converging with the **Chenab River** in **Pakistan**.
- **Ravi River**:
  - Originates from the **Dhauladhar Range** in **Chamba district**, **Himachal Pradesh**.
  - Known for the **Ranjit Sagar Dam**.
- **Sutlej River**:
  - Known as the **Red River**, it rises from **Mansarovar Lake**.
  - Flows south through **Himachal Pradesh**, with significant **hydroelectric** and **irrigation projects** along its course, including the **Bhakra Nangal Dam**.
- **Beas River**:
  - Originates from **Rohtang Pass**, merging with the **Sutlej River** at **Hari-Ke-Pattan** in **Punjab**.

Right-bank Tributaries

The **right-bank tributaries** of the **Indus River** include:

- **Shyok River**
- **Gilgit River**
- **Hunza River**
- **Swat River**
- **Kunnar River**
- **Kurram River**
- **Gomal River**
- **Tochi River**
- **Kabul River**

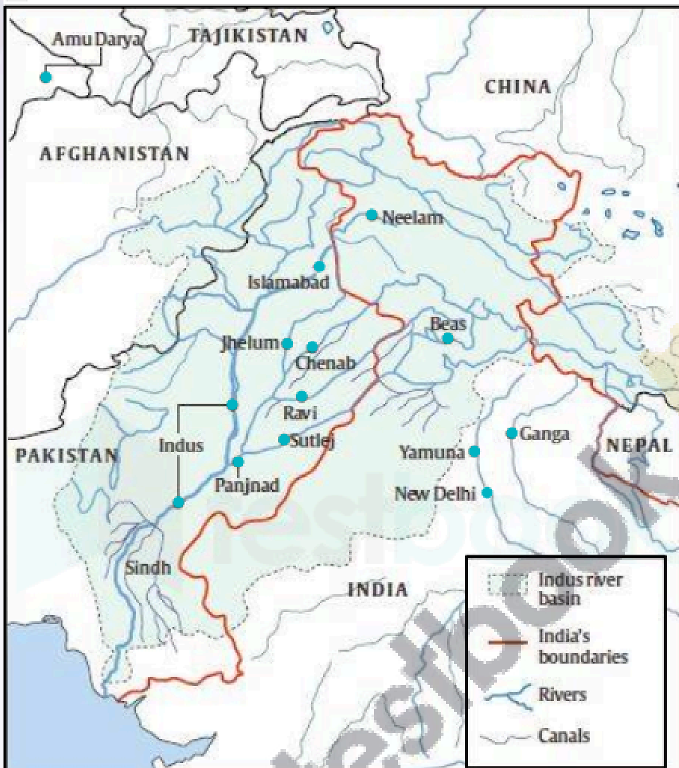
Important Right-bank Tributaries:

- **Shyok River**:
  - Rises from the **Karakoram Range** and flows through **Northern Ladakh**, originating from the **Rimo Glacier**.
- **Nubra River**:
  - A major tributary of the **Shyok River**, originating from the **Nubra Glacier**, it joins the **Shyok River** downstream.

Indus Water Treaty 1960

- The **Indus Water Treaty** (1960) is a landmark **agreement** between **India** and **Pakistan**, brokered by the **World Bank**, aimed at managing the **shared water resources** of the **Indus River system**.
- The treaty allocated the use of the three **eastern rivers—Ravi**, **Beas**, and **Sutlej**—to **India**, while granting **Pakistan** control over the three **western rivers—Indus**, **Jhelum**, and **Chenab**.

- This division of **rivers** established clear **guidelines** for **water-sharing** between the two nations.
- The treaty has been regarded as one of the most **successful water-sharing agreements** globally, providing a framework for **cooperation** and **conflict resolution** between the two countries despite their broader **geopolitical tensions**.
- It allows **India** certain **non-consumptive uses** of the **western rivers**, such as **hydropower projects**, while ensuring a **downstream flow** to **Pakistan**, thereby maintaining a **balance** between the **water needs** and **rights** of both nations.



EASTERN RIVERS	WESTERN RIVERS	TOTALAREA OF INDUS BASIN:
Sutlej. Beas, Ravi 11 cubic km Annual flow before entering Pakistan. India has full rights on all waters. Pakistan can use whatever flows into its territory.	Indus, Jhleum, Chenab 232.5 cubic km Annual flow before entering Pakistan. India can use only about 62.2 cubic km: 170.3 cublic km reserved for Pakistan.	11.65 lakh sq km 47% in Pakistan 39% in India. 8% in China, 6% in Afghanistan.

Ganga River System

The **Ganga River System** is one of the major river systems in **India**, originating from the **Himalayas** and flowing across the northern and eastern parts of the subcontinent. It is crucial for the sustenance of millions, providing water for **agriculture**, **drinking**, and supporting diverse **ecosystems**. This article explores the **origin**, **course**, and **tributaries** of the Ganga River System.

About Ganga River

The **Ganga River** is one of the three major river basins within the **Himalayan Drainage System**. The Ganga and its numerous **tributaries** flow through the northern and eastern parts of India, shaping the landscape and sustaining diverse ecosystems along its course. With a length of over **2,500 kilometers**, the Ganga is the most important river in India and one of the longest rivers in **Asia**.

Origin of Ganga River

The **Ganga River** rises in the **Himalayas**, flowing about **2,525 km** eastward through a vast plain to the **Bay of Bengal**. It crosses five Indian states:

- **Uttarakhand**
- **Uttar Pradesh**
- **Bihar**
- **Jharkhand**
- **West Bengal**

The Ganga has a **catchment area** of **861,404 sq. km**, which is **26.4%** of India's total land area. It drains into the **Bay of Bengal** and is one of the world's most densely populated river basins, housing about **half of India's population**. The river supplies over **one-third** of India's **surface water** and accounts for more than **half** of the country's total **water use**. Beyond its practical significance, the Ganga is also one of India's **holiest rivers**, holding immense **cultural** and **spiritual importance**.

Course of Ganga River

The **Bhagirathi**, considered the **source stream** of the Ganga, emanates from the **Gangotri Glacier** at **Gaumukh**, at an elevation of **3,892 m** (12,770 feet). Other major headstreams include:

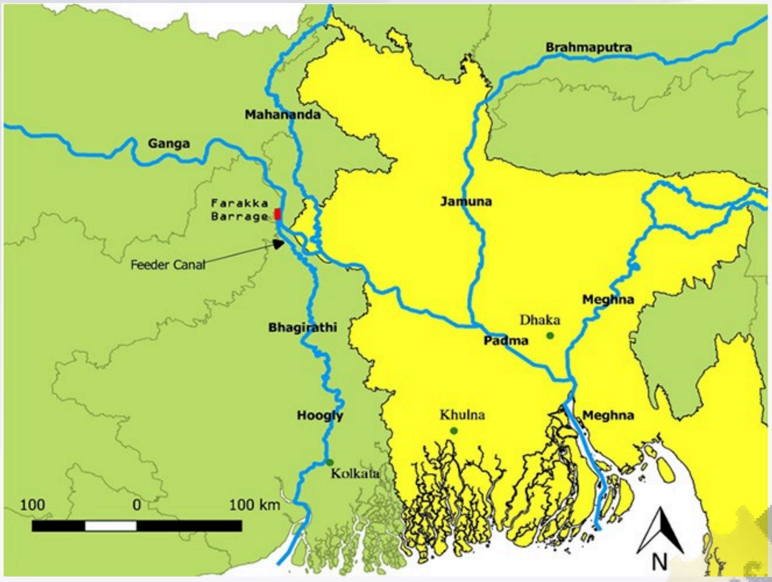
- **Alaknanda**
- **Dhauliganga**
- **Pindar**
- **Mandakini**
- **Bhilangana**

At **Devprayag**, where the **Alaknanda** meets the **Bhagirathi**, the river is named **Ganga**. As the river flows into the **Gangetic Plains** at **Haridwar**, water is diverted into the **Upper Ganga Canal** for **irrigation**. A **barrage** at **Bijnore** directs water into the **Madhya Ganga Canal** during the **monsoon season**, and another at **Narora** diverts water into the **Lower Ganga Canal**. The **Ramganga** joins the Ganga near **Kannauj**, contributing additional water.

The **Yamuna** joins the Ganga at **Prayagraj** (Allahabad), creating a significant confluence. Beyond Prayagraj, the Ganga is joined by several **tributaries** from the north and south. The **Farakka Barrage** in **West Bengal** regulates the flow, diverting a portion of its water into a feeder canal that connects to the **Hooghly River**. Below Farakka, the Ganga divides into two branches:

- **Bhagirathi** (**Hooghly**) on the right
- **Padma** on the left

The **Bhagirathi** (**Hooghly**) reaches the **Bay of Bengal** about **150 km** downstream from **Kolkata**, while the **Padma** enters **Bangladesh** before joining the **Brahmaputra** and **Meghna** rivers and ultimately the **Bay of Bengal**.



Tributaries of Ganga River

Some major **tributaries** of the Ganga River are:

- **Yamuna**
- **Ramganga**
- **Gomti**
- **Ghaghara**
- **Gandak**
- **Damodar**
- **Kosi**

Alaknanda River

The **Alaknanda** is a primary **headstream** of the Ganga, originating at the confluence of the **Satopanth** and **Bhagirath** glaciers in **Uttarakhand**. It joins the **Bhagirathi** at **Devprayag**, where it becomes the Ganga. Its key **tributaries** are:

- **Mandakini**
- **Nandakini**
- **Pindar**

The **Badrinath** pilgrimage center and **Tapt Kund** are located along the **Alaknanda**.

Bhagirathi River

The **Bhagirathi** rises at the **Gangotri Glacier** and converges with the **Alaknanda** at **Devprayag** to form the Ganga. It flows through gorges in the central **Himalayas**.

Dhauliganga River

The **Dhauliganga** originates from **Vasundhara Tal** in **Uttarakhand** and merges with the **Alaknanda** at **Vishnuprayag**. It cuts spectacular **gorges** and has significant **hydroelectric** potential, with projects like the **Tapovan Vishnugad Hydropower Project**.

Ramganga River

The **Ramganga** originates from the southern slopes of **Dudhatoli Hill** in **Chamoli**, **Uttarakhand**, and joins the Ganga near **Kannauj**. It is a major tributary flowing through **Corbett National Park**.

Gomti River

The **Gomti** originates near **Madho Tanda**, **Pilibhit** in **Uttar Pradesh**, and joins the Ganga at **Ghazipur**. The **Sai River** is a significant tributary of the **Gomti**.

Ghaghara River

The **Ghaghara**, also known as **Karnali** or **Kaurial**, originates near **Lake Mansarovar** in **Tibet**. It flows through **Nepal** and enters India, where it merges with the Ganga at **Chhapra**, **Bihar**.

Kosi River

The **Kosi** is an unstable river, often called the **Sorrow of Bihar** due to its frequent floods. It is known as **Saptakoshi** for its seven **tributaries** and joins the Ganga at **Kursela**, **Bihar**.

Gandak River

The **Gandak** originates from the **Kali** and **Trisuli** rivers in the **Himalayas** of **Nepal**, and joins the Ganga at **Sonepur**, near **Patna**.

Cities on the Banks of the Ganga

The Ganga flows through major cities, including:

- **Srinagar**
- **Rishikesh**
- **Haridwar**
- **Roorkee**
- **Bijnor**
- **Narora**

- **Kannauj**
- **Kanpur**
- **Allahabad**
- **Varanasi**
- **Mirzapur**
- **Patna**
- **Bhagalpur**
- **Beharampore**
- **Serampore**
- **Howrah**
- **Kolkata**

Panch Prayag (Confluence Points of the Ganga)

The five **confluence points**, or **Prayag**, of the Ganga system are:

- **Devprayag**: **Alaknanda** and **Bhagirathi**
- **Rudraprayag**: **Mandakini** and **Alaknanda**
- **Nandaprayag**: **Nandakini** and **Alaknanda**
- **Karnaprayag**: **Pindar** and **Alaknanda**
- **Vishnuprayag**: **Dhauliganga** and **Alaknanda**

Ganga-Brahmaputra Delta

Before entering the **Bay of Bengal**, the **Ganga**, along with the **Brahmaputra**, forms the world's largest **delta**. The delta, consisting of **distributaries** and **islands**, is covered with dense **mangroves**. This region is marked by a complex, highly indented **coastline** and low-lying **swamps** that become inundated with **marine water** during high tides.

Brahmaputra River System

The **Brahmaputra River System** is a vital waterway that originates from the **Tibetan Plateau** and flows through the northeastern part of the **Indian subcontinent**. It supports millions by providing water for **agriculture**, **drinking**, and maintaining diverse **ecosystems**. This article explores the **origin**, **course**, and **tributaries** of the Brahmaputra River.

About the Brahmaputra River System

The Brahmaputra is a crucial part of the **Himalayan Drainage System**, one of the three major river basins in the region. Spanning over **2,900 kilometers**, it is one of Asia's longest rivers. The river and its tributaries have a significant role in the **hydrology** and **agriculture** of the areas they flow through.

Origin of the Brahmaputra River System

The **Brahmaputra**, meaning "Son of Brahma," rises from the **Chemayungdung glacier** in **southwestern Tibet**, near the sources of the **Indus** and **Satluj** rivers. Despite its high altitude, the **Tsangpo River**, as it is known in Tibet, flows gently and is navigable for about **640 kilometers**.

Course of the Brahmaputra River System

The river, initially called the **Yarlung Tsangpo** in Tibet, flows through dramatic **gorges** in the **Himalayas**, entering **Arunachal Pradesh** as the **Dihang**. West of **Sadiya**, it is joined by the **Lohit** and **Dibang** rivers and takes the name **Brahmaputra**. Flowing through **Bangladesh** as the **Jamuna River**, it merges with the **Ganga** to form the expansive **Sundarbans Delta**.

Note: The **Majuli** and **Umananda** islands in Assam, located within the Brahmaputra River, are the largest and smallest river islands in the world.

Tributaries of the Brahmaputra River System

Left Bank Tributaries:

1. **Lhasa River**
2. **Nyang River**
3. **Parlung Zangbo River**
4. **Lohit River**
5. **Dhanashri River**
6. **Kolong River**

Right Bank Tributaries:

1. **Kameng River**
2. **Manas River**
3. **Beki River**
4. **Raidak River**
5. **Jaldhaka River**
6. **Teesta River**
7. **Subansiri River**

Key Tributaries Explained:

- **Lohit River**: Originates in **eastern Tibet**, travels through the **Mishmi Hills**, and joins the **Siang** River. Known for its **dense forests** and **medicinal plants**.
- **Subansiri River**: Also known as the **Gold River** due to its gold dust, this river flows through the **Lower Subansiri District** in **Arunachal Pradesh**.

- **Kameng River**: Originates in **Tawang**, flows through **West Kameng** and **Sonitpur districts**, and is near **Kaziranga National Park**.
- **Manas River**: A **transboundary** river that originates in **Bhutan**, flows through **Assam**, and merges with the Brahmaputra at **Jogighopa**. The valley is home to protected areas like **Royal Manas National Park**.
- **Teesta River**: Begins at **Tso Lhamo Lake** in **North Sikkim** and converges with the **Rangeet River** at **Tribeni**.
- **Dibang River**: Originates from the southern slopes of the **Himalayas**, flowing into the plains of **Arunachal Pradesh** near **Nizamghat**.
- **Kopili River**: The largest south-bank tributary in **Assam**, flowing through **Meghalaya** and **Assam**, threatened by **hydroelectric projects** and **pollution** from coal mining.

Different Names of the Brahmaputra River:

- **Tibet**: Tsangpo (The Purifier)
- **China**: Yarlung Zangbo, Jiangin
- **Assam Valley**: Dihang or Siang, later Brahmaputra
- **Bangladesh**: Jamuna River, later Padma after merging with the **Ganga**

States the Brahmaputra Flows Through:

- **Arunachal Pradesh**
- **Assam**
- **Meghalaya**
- **Nagaland**
- **West Bengal**
- **Sikkim**

Cities on the Brahmaputra River:

- **Dibrugarh**
- **Pasighat**
- **Neamati**
- **Tezpur**
- **Guwahati** (major urban center)

Hydel Power Projects on the Brahmaputra River System:

- **Arunachal Pradesh**: Tawang, **Subansiri**, **Ranganadi**, **Paki**, **Papumpap**, **Kameng**.
- **Sikkim**: Rangit, **Tista**.
- **Assam**: **Kopili**.
- **Meghalaya**: New **Umtru**.
- **Nagaland**: **Doyang**.
- **Manipur**: **Loktak**, **Tipaimukh**.
- **Mizoram**: **Tuibai**, **Tuirial**, **Dhaleshwari**.